

Instructor Effectiveness Analysis

A Data-Driven Study of Student Engagement & Learning Outcomes

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Executive Summary

This report analyzes 2,000 batch-level education records to identify what drives instructor effectiveness and student success. The analysis finds that completion rate is the single strongest lever available: it is very strongly and negatively correlated with dropout (-0.95), and instructors with higher completion rates consistently receive better feedback and produce better academic outcomes. The report translates these findings into eight concrete, prioritized recommendations for improving student engagement and instructor performance.

Business Problem

Educational platforms invest heavily in instructors, but without measurable performance data it is difficult to know which instructors and courses are driving strong outcomes, and which are causing students to disengage or drop out. This analysis was undertaken to answer three questions:

- Which factors are most strongly linked to student dropout and low completion?
- Which instructors and courses are performing best, and why?
- What concrete actions can improve completion rate, engagement, and satisfaction?

Dataset & Methodology

The dataset contains 2,000 batch-level records across instructors, courses, and batches, with the following fields: completion rate, average score improvement, average quiz score, dropout rate, average watch time, assignment submission rate, forum activity rate, average feedback score, and feedback response rate.

Data Cleaning

- Checked for missing values and duplicate records
- Verified data types and valid percentage ranges
- Inspected numerical columns for outliers using boxplots

Analysis Steps

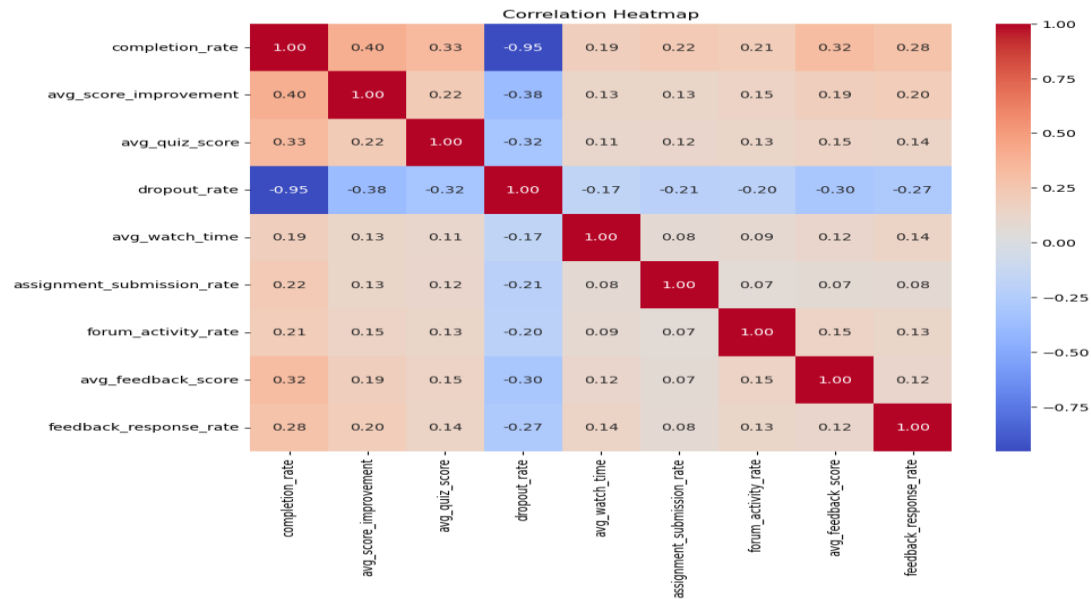
- Exploratory Data Analysis (distributions, summary statistics)
- Correlation analysis across all numerical metrics
- Relationship analysis between engagement and performance metrics
- Group-wise ranking of instructors, courses, and batches

Key Findings

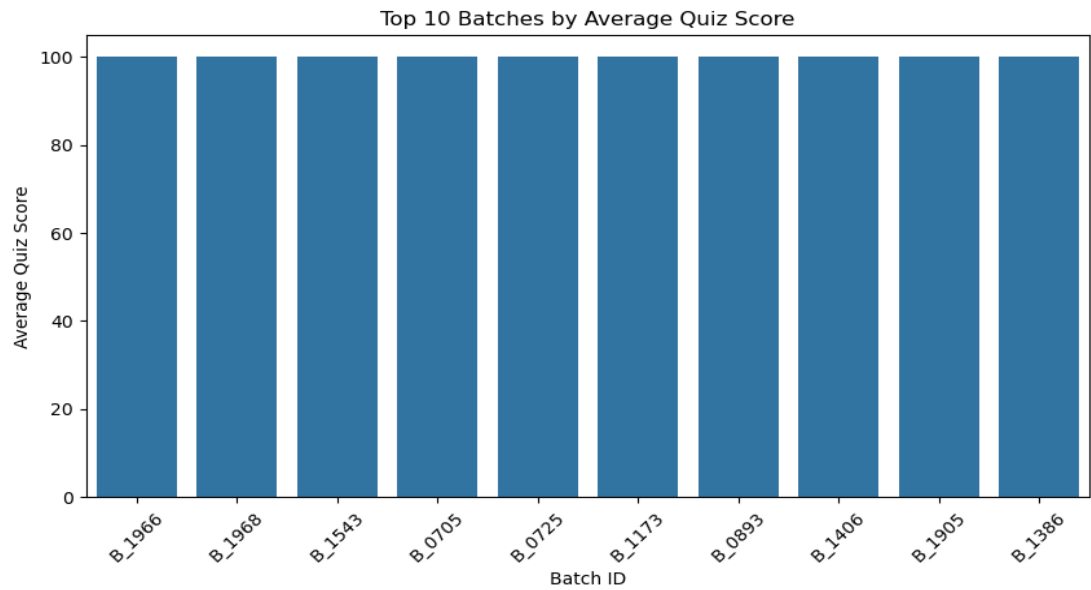
Metric Relationship	Finding
Completion rate vs. Dropout rate	Very strong negative correlation (-0.95)
Completion vs. Quiz score	Course completers score meaningfully higher

Completion vs. Feedback score	Higher completion instructors get better feedback
Watch time vs. Quiz score	Only weak positive correlation
Assignment submission vs. Completion	Positively related — engagement matters
Forum activity vs. Score improvement	Weak positive association

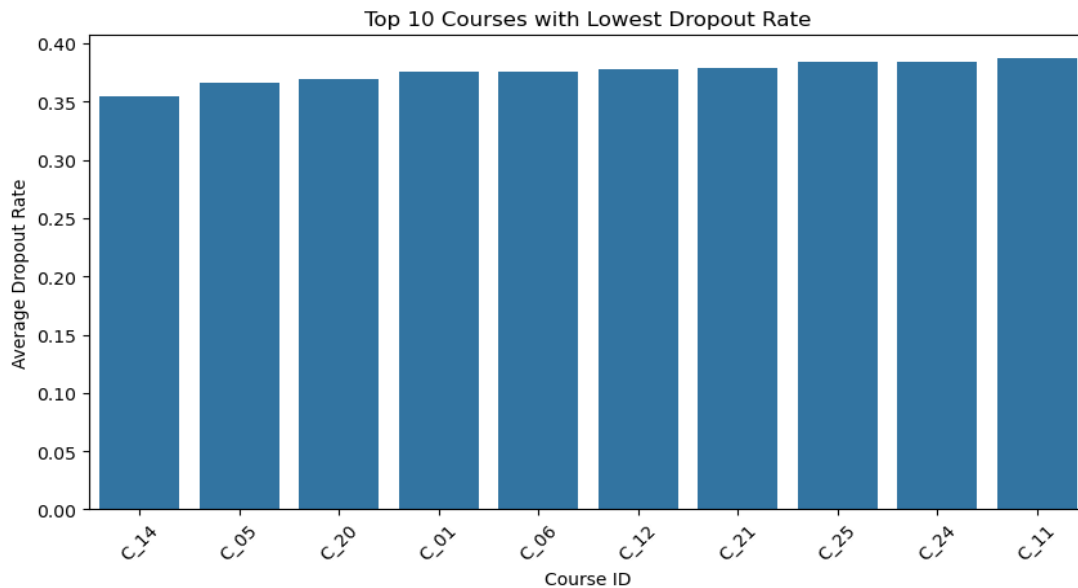
Correlation Between Metrics



Top 10 Batches by Average Quiz Score



Courses with the Lowest Dropout Rates



Recommendations

- Encourage instructors to adopt teaching strategies used by the highest-performing instructors.
- Monitor courses with high dropout rates and identify the causes of student disengagement.
- Increase assignment participation through reminders and structured deadlines.
- Promote discussion forum participation to improve student interaction and collaborative learning.
- Encourage instructors to respond promptly to student feedback to enhance satisfaction.
- Regularly monitor instructor performance using completion rate, feedback score, and quiz performance as KPIs.
- Use data-driven decision-making to identify underperforming courses and provide targeted support.

Conclusion

This analysis shows that completion rate is the central driver of student outcomes: it is strongly associated with lower dropout and better performance across every metric examined. Correlation analysis, visualizations, and group-wise comparisons identified clear performance patterns among instructors, courses, and batches. These findings give educational institutions a data-driven basis for improving teaching quality, student engagement, and learning effectiveness.

Future Scope

- Develop a machine learning model to predict instructor effectiveness.
- Build an interactive dashboard using Tableau or Power BI.
- Analyze instructor performance trends across multiple semesters.
- Incorporate student demographic and attendance data.
- Automate performance reporting for instructors and management.